HDOC Day-56 | C Programming

Problem: Read an integer array and window size k. Pass the array by pointer to first_negative(). For each contiguous window of length k, print its first negative number, or 0 if none exists. Print -1 for an invalid k. Use nested loops.

C Program

```
#include <stdio.h>

void first_negative(int *arr, int size, int k) {
    if (k <= 0 || k > size) {
        printf("-1\n");
        return;
    }

    for (int i = 0; i <= size - k; i++) {
        int first = 0;
        for (int j = i; j < i + k; j++) {
            if (*(arr + j) < 0) {
                first = *(arr + j);
                break;
            }
        }
        if (i > 0) printf(" ");
        printf("%d", first);
    }
    printf("\n");
}

int main(void) {
    int size, k;
    scanf("%d", &size);
    if (size <= 0) return 0;
    int arr[size];
    for (int i = 0; i < size; i++)
        scanf("%d", &arr[i]);
    scanf("%d", &k);
    first_negative(arr, size, k);
    return 0;
}
```

Sample Input and Output

```
Input:
7
12 -1 -7 8 -15 30 16
3

Output:
-1 -1 -7 -15 -15
```

Explanation: The outer loop visits each window. The inner loop checks values from left to right and stops at the first negative. A window with no negative prints 0. Valid k is from 1 to size.